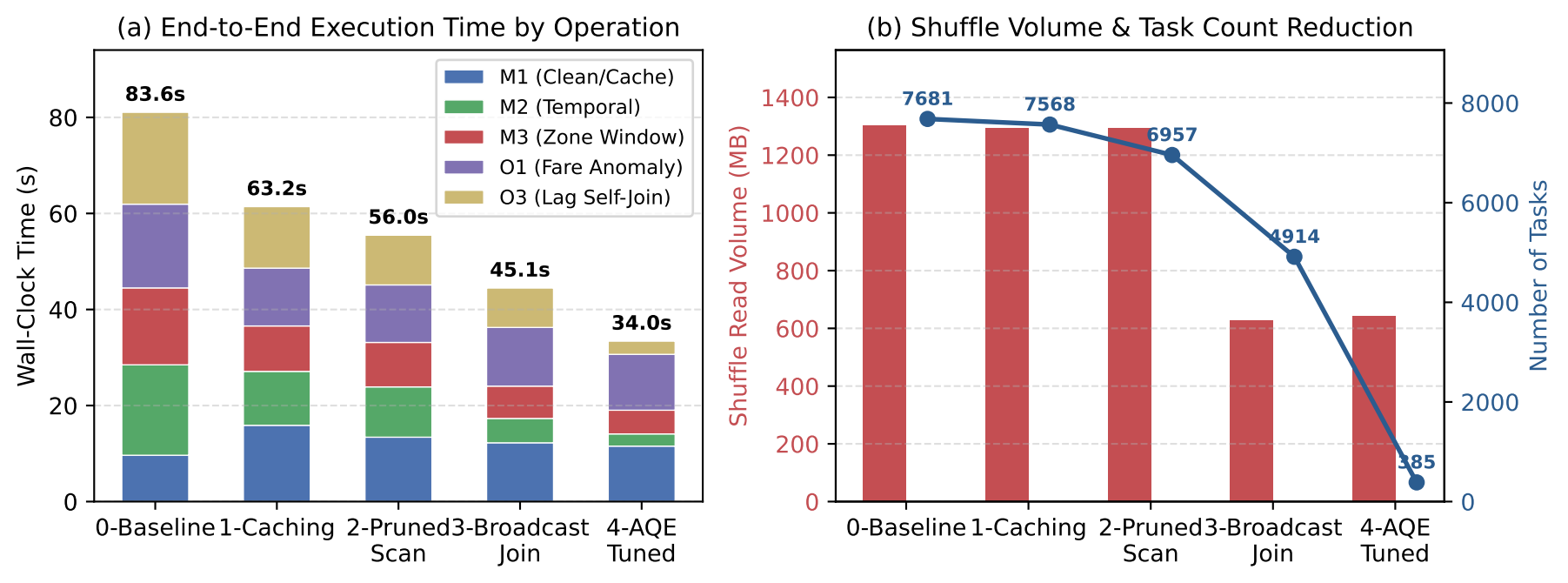
(a) End-to-End Execution Time by Operation
Wall-Clock Time (s)
M1 (Clean/Cache)
M2 (Temporal)
M3 (Zone Window)
O1 (Fare Anomaly)
O3 (Lag Self-Join)
83.6s
63.2s
56.0s
45.1s
34.0s
0-Baseline
1-Caching
2-Pruned Scan
3-Broadcast Join
4-AQE Tuned
(b) Shuffle Volume & Task Count Reduction
Shuffle Read Volume (MB)
Number of Tasks
7681
7568
6957
4914
385